

A literature-implied Mazur–Ulam subcase for numerical-index-one spaces

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Status

literature_implied_answer (partial subcase).

Source question

The source paper is Dongni Tan, Xujian Huang, and Rui Liu, *Generalized-Lush Spaces and the Mazur-Ulam Property*, arXiv:1210.7324. After proving that every local-GL space, hence every lush space, has the Mazur–Ulam property, the authors ask at the end of the paper:

Does every Banach space with numerical index 1 have the MUP?

Supporting theorem

The supporting source is Vladimir Kadets, Miguel Martin, Javier Meri, and Antonio Perez, *Spear operators between Banach spaces*, arXiv:1701.02977. In the identity case, that source records that X has numerical index one exactly when Id_X is a spear operator; numerical index one implies the alternative Daugavet property (aDP). It also cites the SCD theorem that if X has the aDP and B_Y is SCD for every separable subspace $Y \subset X$, then X is lush. In particular, this applies when X has the convex point of continuity property, the Radon–Nikodym property, or contains no copy of ℓ_1 .

Deduction

Let X be a Banach space with numerical index one and suppose that B_Y is SCD for every separable subspace $Y \subset X$. Since $n(X) = 1$, Id_X is spear, so X has the aDP. The SCD theorem from Kadets–Martin–Meri–Perez then implies that X is lush. By the main corollary of Tan–Huang–Liu, every lush space has the Mazur–Ulam property. Therefore X has the MUP.

Thus the final question of arXiv:1210.7324 has an affirmative answer for numerical-index-one spaces satisfying the SCD separable-subspace condition. This includes finite-dimensional spaces and, more generally, spaces with CPCP, RNP, or no isomorphic copy of ℓ_1 .

Scope

This is not a full solution of the unrestricted numerical-index-one question. The implication still goes through lushness, while both the source paper and the later spear-operator literature record numerical-index-one spaces which are not lush. The remaining case is precisely outside this SCD/lushness mechanism.

Search notes

Local run indexes were searched for arXiv:1210.7324 and the core phrases “Mazur–Ulam”, “MUP”, “Tingley”, “numerical index”, “lush”, “local-GL”, and “spear”. Web searches on 2026-06-29 for combinations of “numerical index one”, “Mazur–Ulam property”, “MUP”, “Tingley’s problem”, “lush”, and “SCD” found no exact full answer to the unrestricted question. The implication recorded here is agent-identified from the two papers above.

References

- [1] D. Tan, X. Huang, and R. Liu, *Generalized-Lush Spaces and the Mazur-Ulam Property*, arXiv:1210.7324.
- [2] V. Kadets, M. Martin, J. Meri, and A. Perez, *Spear operators between Banach spaces*, arXiv:1701.02977; Lecture Notes in Mathematics 2205, Springer, 2018.
- [3] T. Banach, *Every 2-dimensional Banach space has the Mazur-Ulam property*, arXiv:2103.09268; Linear Algebra Appl. 632 (2022), 268–280.